UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 24347

LH EVOLUTION OF COSMIC STRUCTURE

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Consider a flat universe with $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_{\mathrm{M},0} = 0.3$, $\Omega_{\Lambda,0}$ = 0.7. In the Press-Schechter formalism the comoving density, $\frac{dn}{d\log M}$, of halos of mass M is

$$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_M}{M}\left|\frac{n_{\mathrm{eff}}+3}{6}\right|\nu e^{-\nu^2/2},$$

where ν is defined as $\nu = 1.69/\sigma_{\mathrm{M}}$, and the matter density fluctuation (σ_M) on mass scale (M) is $\sigma_{\mathrm{M}} \propto M^{-(n_{\mathrm{eff}}+3)/6}$. n_{eff} is the effective spectral index, and ρ_{M} is the comoving matter density. Calculate the comoving density of halos at redshift $z = 0$ with mass of 1×10^{15} $\mathrm{M}_\odot$ in a universe where $\sigma_8 = 0.85$ and $n_{\mathrm{eff}} = -1.6$. **[10]**

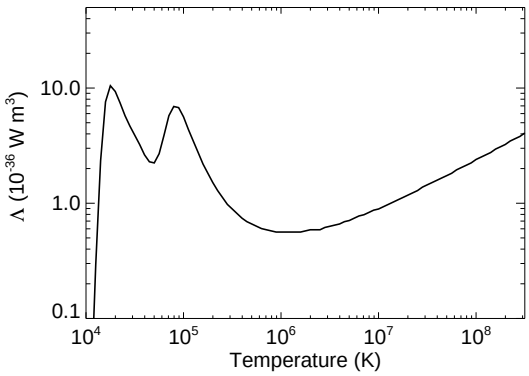
(b) In galaxy clusters, the observed relation between X-ray luminosity (L_X) and gas temperature (T) is $L_X \propto T^3$. Assuming the cluster is virialised, show that the expected relation for self-similar clusters is $L_X \propto T^2$. If one wanted to explain this difference as arising from a systematic trend in the gas fraction (f_{gas}) with temperature, calculate the relation between f_{gas} and T that would be required? **[12]**

(c) Consider a spherically-symmetric dark matter halo filled with hot gas in hydrostatic equilibrium. Assume a gas temperature profile of T, a gas density profile of ρ_{gas} and that the gas obeys the ideal gas law. Show that the enclosed mass of the halo is

$$M(r) = -\frac{k_{\mathrm{B}}Tr}{\mu m_{\mathrm{H}}G}\left[\frac{\mathrm{dln}\rho_{\mathrm{gas}}}{\mathrm{dln}r} + \frac{\mathrm{dln}T}{\mathrm{dln}r}\right],$$

where μ is the mean molecular weight in units of the Hydrogen mass m_{H}. **[8]**

2. (a) The figure below shows the *cooling function* (Λ) of a metal-free gas as a function of the temperature of the gas. Describe the origin of the peak at $\sim 2 \times 10^4$ K , the peak at $\sim 10^5$ K and the power law at temperatures greater than 10^7 K.



[6]

(b) A cluster of virial mass 10^{15} $\mathrm{M}_\odot$ is observed at redshift $z = 0.5$. Calculate the radius of the virialised region of the cluster at this redshift. Assume $\Omega_{\Lambda,0} = 0.7$, $\Omega_{M,0} = 0.3$ and $H_0 = 70\ \mathrm{km\ s^{-1} Mpc^{-1}}$. **[6]**

(c) Assume that the mass density of a galaxy cluster is given by a Navarro-Frenk-White (NFW) profile in which the scale radius, R_S, is $R_S = R_{\mathrm{virial}}/3$. In this cluster, the gas density (ρ_g) is related to its mass density (ρ_m) as $\rho_g = 0.17\rho_m$ and the central mass density ($\rho_{m,0}$) is $6 \times 10^4\ \mathrm{amu\ m^{-3}}$. The gas is at a uniform temperature of 10^7 K. Using the cooling function in part (a), calculate the cooling time at $r = 0.05R_{\mathrm{virial}}$ and $r = 0.75R_{\mathrm{virial}}$. **[12]**

(d) Observations show that galaxies at the center of galaxy clusters are not actively forming stars. What process is likely needed to reconcile this observation with the cooling times calculated in part (c)? What observational evidence is there for such a process? **[6]**

3. (a) Derive an expression for the Fundamental Plane of elliptical galaxies, which relates the radius of a galaxy (R) to its surface brightess (I) and the velocity dispersion of the stars within it (σ). You may assume the galaxies obey the Virial Theorem and have a mass-to-light ratio (M/L), which scales as $M/L \propto \sigma^{0.8}$.
Give one reason why galaxies might have an M/L ratio that scales with σ in this way. **[12]**

(b) Consider a universe in which the baryonic Jeans mass immediately following decoupling is $1.2 \times 10^5\ \mathrm{M}_\odot$. If the ratio of the sound speeds after decoupling and before decoupling is 2.6×10^{-5}, calculate the baryonic Jeans mass just prior to decoupling.
Comment on what these values mean for the formation of structure. **[6]**

(c) The two-point correlation function of galaxies $\xi(r)$ is defined such that the expected number of galaxies within a volume V at a distance R from a particular galaxy is $\langle N_g \rangle = n[1 + \xi(r)]V$, where n is the mean number-density of galaxies. Calculate the form of $\xi(r)$ if all galaxies are distributed uniformly on (i) a single sheet or (ii) a filament.
Explain qualitatively how the correlation function would be modified if there are $multiple$ sheets or filaments wrapped around voids with characteristic radii R. **[12]**

Formula Sheet

Friedmann equation
$H^2 = \frac{8}{3}\pi G\rho - \frac{k}{a^2}$

Fluid equation
$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + P\right) = 0$

Acceleration equation
$\frac{\ddot{a}}{a} = -\frac{4}{3}\pi G\left(\rho + 3P\right)$

Equation of State
$P = \omega\rho$

Critical density
$\rho_c = \frac{3H^2}{8\pi G}$

Hubble parameter
$H = \frac{\dot{a}}{a}$

$$H(z) = H_0\sqrt{\frac{\Omega_{M,0}}{a^3} + \frac{\Omega_{r,0}}{a^4} + \Omega_{\lambda,0}}$$

Scale factor (Matter only)
$a(t) \propto t^{2/3}$

Scale factor (Radiation only)
$a(t) \propto t^{1/2}$

Overdensity
$\delta(r,t) = \frac{\rho(r,t) - \bar{\rho}(t)}{\bar{\rho}(t)}$

Dynamical time (gravitational)
$t_{dyn} = \frac{1}{\sqrt{4\pi G\bar{\rho}}}$

Press-Schechter number density
$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\rm M}}{M}\left|\frac{n_{\rm eff}+3}{6}\right|\nu e^{-\nu^2/2}$

NFW density profile
$\rho(r) = \frac{\rho_0}{(r/r_s)(1+r/r_s)^2}$

Optical depth
$\tau_e = \int n_e \sigma_T dx$

Feedback energy from supernova
$E_{fb} = \epsilon_{SN}\,\xi\,M_*\,E_{SN}$

Feedback energy from black holes
$E_{AGN} = \epsilon_{BH}\,M_{BH}\,c^2$

Radiation pressure
$P_{rad} = \frac{L}{c}\frac{1}{4\pi R^2}$

Ram pressure condition
$\rho_{icm}v^2 = 2\pi G\Sigma(R)\Sigma_g(R)$

X-ray luminosity
$L_x \propto \rho^2\,V\Lambda(T)$

Correlation function
$\langle N_g\rangle = n[1 + \xi(r)]V$

Radius of spherical top hat model
$R(t) = \frac{R_*}{2}\left(\frac{6t}{\alpha R_*}\right)^{2/3}\left[1 - \frac{1}{20}\left(\frac{6t}{\alpha R_*}\right)^{2/3}\right]$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.